

Name: Kevin 

Date: 11/3/2024

Instructor: Samuel Chukwuemeka

Project: Federal Income Tax Rates

Company Name: Forbes

Website: <https://www.forbes.com/sites/kellyphillipserb/2023/11/09/irs-announces-2024-tax-brackets-standard-deductions-and-other-inflation-adjustments/>

Objectives:

For the Single Filing Option:

- (1.) Calculate the federal income taxes of any United States resident or citizen within each taxable income bracket manually using Arithmetic method for year 2024.
- (2.) Write a piecewise function of the federal income tax rates for the year 2024.

(3.) Recalculate the same income tax of the United States resident or citizen for those income brackets algebraically using the Piecewise Function method for the year 2024.

Information:

FEDERAL TAX RATES 2024

If taxable income is (\$):	The Tax Due (\$):
0-\$11,600	10% of taxable income
\$11,601-\$47,150	\$1160 + 12% of the amount over \$11,600
\$47,151-\$100,525	\$5426 + 22% of the amount over \$47,150
\$100,526-\$191,950	\$17168.50 + 24% of the amount over \$100,525
\$191,951-\$243,725	\$39,110.50 + 32% of the amount over \$191,950
\$243,726-\$609,350	\$55,678.50 + 35% of the amount over \$243,725
\$609,351+	\$183,647.25 + 37% of the amount over \$609,350

Conversions:

$$10\% = .10$$

$$12\% = .12$$

$$22\% = .22$$

$$24\% = .24$$

$$32\% = .32$$

$$35\% = .35$$

$$37\% = .37$$

Amount (\$) of Taxable Income to test:

I will test the following Taxable Income Amounts (\$):

(1.) \$10,000

(2.) \$26,379

(3.) \$51,004

(4.) \$127,984

(5.) \$240,365

(6.) \$526,375

(7.) \$714,112

Arithmetic Method:

- (1.) For the taxable income of 10,000:

$$Tax = 10000.00 * .10$$

$$Tax = \$1000$$

- (2.) For the taxable income of 26,379:

$$Tax = 1160.00 + .12 (26379 - 11600)$$

$$Tax = 1160.00 + .12(14779)$$

$$Tax = 1160.00 + 1773.48$$

$$Tax = \$2933.48 \cong \$2933$$

- (3.) For the taxable income of 51,004:

$$Tax = 5426 + .22(51004 - 47150)$$

$$Tax = 5426 + .22(3854)$$

$$Tax = 5426 + 847.88$$

$$Tax = \$6273.88 \cong \$6274$$

(4.) For the taxable income of 127,984:

$$Tax = 17168.50 + .24 (127984 - 100525)$$

$$Tax = 17168.50 + .24(27459)$$

$$Tax = 17168.50 + 6590.16$$

$$Tax = \$23758.66 \cong \$23759$$

(5.) For the taxable income of 240,365:

$$Tax = 39110.50 + .32 (240365 - 191150)$$

$$Tax = 39110.50 + .32(48415)$$

$$Tax = 39110.50 + 15492.80$$

$$Tax = \$54603.30 \cong \$54603$$

(6.) For the taxable income of 526,375:

$$Tax = 55678.50 + .35 (526375 - 243725)$$

$$Tax = 55678.50 + .35 (282650)$$

$$Tax = 55678.50 + 98927.50$$

$$Tax = \$154606$$

(7.) For the taxable income of 714,112:

$$Tax = 183647.25 + .37 (714112 - 609350)$$

$$Tax = 183647.25 + .37 (104762)$$

$$Tax = 183647.25 + 38761.94$$

$$Tax = \$222409.19 \cong \$222409$$

Piecewise Function:

Let us define variables that we shall use:

Let: t be the federal taxable income in \$

$f(t)$ is the tax due to federal government in \$

Federal Taxable Income Tiers for 2024

First Piece: (First Tier): First 11,600: $0 \leq t \leq 11600$

$$f(t) = .10t$$

Second Piece: (Second Tier): From 11601- 47150:

$$11601 \leq t \leq 47150$$

$$f(t) = 1160 + .12(t - 11600)$$

$$f(t) = 1160 + .12t - 1392$$

$$f(t) = 0.12t - 232$$

Third Piece: (Third Tier): From 47151- 100525:

$$47151 \leq t \leq 100525$$

$$f(t) = 5426 + .22(t - 47150)$$

$$f(t) = 5426 + .22t - 10373$$

$$f(t) = 0.22t - 4947$$

Fourth Piece: (Fourth Tier): From 100526- 191950:

$$100526 \leq t \leq 191950$$

$$f(t) = 17168.50 + .24(t - 100525)$$

$$f(t) = 17168.50 + .24t - 24126$$

$$f(t) = 0.24t - 6957.50$$

Fifth Piece: (Fifth Tier): From 191950- 243725:

$$191951 \leq t \leq 243725$$

$$f(t) = 39110.50 + .32(t - 191950)$$

$$f(t) = 39110.50 + .32t - 61424$$

$$f(t) = 0.32t - 22313.50$$

Sixth Piece: (Sixth Tier): From 243726- 609350:

$$243726 \leq t \leq 609350$$

$$f(t) = 55678.50 + .35(t - 243725)$$

$$f(t) = 55678.50 + .35t - 85303.75$$

$$f(t) = 0.35t - 29625.25$$

Seventh Piece: (Seventh Tier): Above 609351: $609351 \leq t$

$$f(t) = 183647.25 + .37(t - 609350)$$

$$f(t) = 183647.25 + .37t - 225459.50$$

$$f(t) = 0.37t - 41812.25$$

$$f(t) = \begin{cases} .10t; & 0 \leq t \leq 11600 \\ .12t - 232; & 11601 \leq t \leq 47150 \\ .22t - 4947; & 47151 \leq t \leq 100525 \\ .24t - 6957.50; & 100526 \leq t \leq 191150 \\ .32t - 22313.50; & 191151 \leq t \leq 243725 \\ .35t - 29625.25; & 243726 \leq t \leq 609350 \\ .37t - 41812.25; & 609351 < t \end{cases}$$

Piecewise Function Method (Algebraic Method):

I will test the same numbers as I did with the Arithmetic Method.

(1.) For the Federal taxable wage of 10,000:

10000 falls in the first piece

$$f(t) = 0.10t$$

$$f(10000) = 0.10(10000)$$

$$f(10000) = \$1000$$

(2.) For the Federal taxable wage of 26,379:

26379 falls in the second piece

$$f(t) = .12t - 232$$

$$f(26379) = 0.12(26379) - 232$$

$$f(26379) = 3165.48 - 232$$

$$f(26379) = \$2933.48 \cong \$2933$$

(3.) For the Federal taxable wage of 51,004:

51004 falls in the third piece

$$f(t) = 0.22t - 4947$$

$$f(51004) = 0.22(51004) - 4947$$

$$f(51004) = 11220.88 - 4947$$

$$f(51004) = \$6273.88 \cong \$6274$$

- (4.) For the Federal taxable wage of 127,984:

127984 falls in the fourth piece

$$f(t) = 0.24t - 6957.50$$

$$f(127984) = 0.24(127984) - 6957.50$$

$$f(127984) = 30716.16 - 6957.50$$

$$f(127984) = \$23758.66 \cong \$23759$$

- (5.) For the Federal taxable wage of 240,365:

240365 falls in the fifth piece

$$f(t) = 0.32t - 22313.50$$

$$f(240365) = 0.32(240365) - 22313.50$$

$$f(240365) = 76916.80 - 22313.50$$

$$f(240365) = \$54603.30 \cong \$54603$$

- (6.) For the Federal taxable wage of 526,375:

526375 falls in the sixth piece

$$f(t) = 0.35t - 29625.25$$

$$f(526375) = 0.35(526375) - 29625.25$$

$$f(526375) = 184231.25 - 29625.25$$

$$f(526375) = \$154606$$

(7.) For the Federal taxable wage of 714,112:

714112 falls in seventh piece

$$f(t) = 0.37t - 41812.25$$

$$f(714112) = 0.37(714112) - 41812.25$$

$$f(714112) = 264221.44 - 41812.25$$

$$f(714112) = \$222409.19 \cong \$222409$$

The exact results of the numbers tested using the Arithmetic Method and the Algebraic Method are the same.

Works Cited (MLA):

Chukwuemeka, Samuel. "Piecewise Function Application: Federal Tax."

Appspot.com, 2018,

conferencepresentations.appspot.com/Projects/PreCalculus/PiecewiseFunctions/FederalTax.html#projectRequirements. Accessed 12 Nov. 2024.

Erb, Kelly Phillips. "IRS Announces 2024 Tax Brackets, Standard Deductions and

Other Inflation Adjustments." *Forbes*, 9 Nov. 2023,

www.forbes.com/sites/kellyphillipserb/2023/11/09/irs-announces-2024-tax-brackets-standard-deductions-and-other-inflation-adjustments/.